MIL-T-82009B(YD)
18 May 1984
SUPERSEDING
MIL-T-82009A(YD)
19 December 1977

MILITARY SPECIFICATION

TONGS, MATERIAL LIFTING, PIPE

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two classes of metal tongs for lifting materials such as pipe and conduit.

1.2 Classification. Tongs shall be one of the classes and sizes listed below, as specified (see 6.2). When specified, liners for tongs shall be provided for handling steel, cast iron, or other types of pipe of the outside diameter (od) as specified (see 6.2).

Class I - Cast aluminum.
Class II - Steel.

Size 4 - For use with 4-inch diameter asbestos-cement pipe.
Size 6 - For use with 6-inch diameter asbestos-cement pipe.
Size 8 - For use with 8-inch diameter asbestos-cement pipe.
Size 10 - For use with 10-inch diameter asbestos-cement pipe.
Size 12 - For use with 12-inch diameter asbestos-cement pipe.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3940

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SPECIFICATIONS

FEDERAL

QQ-B-637 - Brass, Naval: Rod, Wire, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strip).

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL STD-129 - Marking for Shipment and Storage.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Specification for Structural Steel.
A148 - Specifications for High-Strength Steel Castings for Structural Purposes.
A242 - Specification for High Strength Low Alloy Structural Steel.
A668 - Specifications for Steel Forgings, Carbon and Alloy, for General Industrial Use.
B21 - Specification for Naval Brass Rod, Bar, and Shapes.
B179 - Specification for Aluminum-Base Alloys in Ingot Form for Sand Castings, Die Castings, and Permanent Mold Castings.
B247 - Specification for Aluminum-Alloy Die and Hand Forgings (Metric).
D3951- Commercial Packaging, Standard Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Design and construction. Tongs shall be of the lever, double jaw, open caliper type with compound leverage grip. The jaws or liners shall grip the outside of the pipe at a point below the centerline of the pipe. The tongs shall be constructed to provide a nonmanual positive release from the pipe after the pipe is set down, and the hoisting line is slacked off. Tongs shall be of a configuration suitable for operation in a class side clearance for handling pipe in a narrow trench. The design and construction of the tongs shall provide adequate strength and grip of cast or fabricated parts to meet the performance requirements represented by the tests in 4.6. Class I tongs shall be of heat-treated aluminum alloy as specified in 3.3.1; and shackles, links, and rings shall be aluminum-allow forgings as specified in 3.3.2; or shackles shall be of cast-aluminum alloy as specified in 3.3.1. Class II tongs shall be fabricated of steel plate as specified in 3.3.3, or cast steel as specified in 3.3.4; shackles, links, and rings shall be steel forgings as specified in 3.3.5; or shackles shall be steel castings as specified in 3.3.4. Fulcrum pins shall be brass as specified in 3.3.6.

3.2 First article. When specified (see 6.2), the contractor shall furnish one tong for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.3.1 Cast aluminum. Cast aluminum parts shall be aluminum-alloy castings, heat treated, conforming or equivalent to ASTM B179. The alloy designation shall be that used by the manufacturer in regular commercial production of the parts.

3.3.2 Aluminum forgings. Aluminum forging parts shall be aluminum-alloy forgings conforming or equivalent to ASTM B247. The alloy designation shall be that used by the manufacturer in regular commercial production of the parts.

3.3.3 Steel. Steel plates, shapes, and bars shall conform to ASTM A36 or A242, as applicable, and as standard with the manufacturer.

3.3.4 Steel castings. Steel castings shall conform to ASTM A148.

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3.3.5 Steel forgings. Steel forgings shall conform to ASTM A668.

3.3.6 Brass. Brass bars and pins shall conform to QQ-B-637, composition 1; or to ASTM B21, alloy A.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.6 Workmanship. The quality of workmanship shall meet the standards prevalent among manufacturers who normally produce equipment of the type specified herein.

3.6.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.6.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.6.3 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.6.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk

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or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.6.5 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and otherwise to the standards of good commercial practice.

3.6.6 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.6.7 Forgings. All forgings shall be of uniform quality, free of detrimental flash, scale, cracks, hard spots, cold shuts, and excessive cooling stresses. Strength and other essential physical properties of the forgings shall be adequate throughout the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one tong when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

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4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All tongs offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level shall be level II and the Acceptable Quality Level shall be 4.0 percent defective. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.4 Examination. Each sample selected shall be examined for compliance with the requirements in section 3 of this specification. This element of inspection shall encompass all visual examinations and dimensional measurements.

4.5 Tests.

* 4.5.1 Operational weight test. The sample tongs shall be set manually at the center of a weighted pipe which shall be loaded by any convenient means to obtain a gross load on the tongs for the size specified in 1.2. Gross loads shall be not less than the following:

Size 4 - 1,250 pounds (lb).
Size 6 - 1,800 lb.
Size 8 - 2,500 lb.
Size 10 - 4,000 lb.
Size 12 - 4,500 lb.

The weighted pipe represents the heaviest load to be handled allowing a safety factor of not less than two. The tongs and weighted pipe shall be lifted, then swung in an arc and set down. The tongs shall be released from the pipe by slacking off of the line without any manual manipulation of the tongs. Failure of the tongs to pass any test shall be cause for rejection (see 3.1).

4.6 Packaging inspection. The preservation, packing, and marking of the tongs shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

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5.1.1.2 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.3 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.3.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.3.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with waterproof tape.

5.1.1.4 Unit protection. The complete tong shall be preserved method I or III as applicable.

5.1.2 Commercial. Material shall be preserved in accordance with ASTM D3951.

* 5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Levels A and B. Packing shall be in accordance with MIL-STD-794. Containers shall be selected from table I of MIL-STD-794 for the appropriate level.

5.2.2 Commercial. Material shall be packed in accordance with ASTM D3951.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Material lifting tongs are lifting devices intended to be used for lifting and moving pipe and conduit.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class and size of tongs required (see 1.2).
c. When liners for tongs are required, and the od of the pipe to be handled (see 1.2 and 6.5).
d. Specifications and standards (see 2.1.1).
e. When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
f. Color of finish coat (see 3.5).
g. Level of preservation and level of packing required (see 5.1 and 5.2).

* 6.3 First article. When a first article inspection is required, (see 3.2 and 6.2) the item will be tested and should be a sample selected from the first

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run production items or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.5 General information. When ordering tongs for handling material other than cement-asbestos pipe, the od of pipe, such as steel, cast iron, and concrete, or other cylindrical material must be specified in order that the proper liners or spacer pads be furnished (see 6.2).

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-82009B (YD)	TONGS, MATERIAL LIFTING, PIPE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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